

Predicting Real Estate ROI Using Socioeconomic, Educational, and Interest Rate Data: A Multi-Window Historical Analysis

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Abstract

This paper investigates whether neighborhood-level socioeconomic indicators and historical mortgage interest rate data can predict 3-year residential real estate Return on Investment (ROI) across U.S. zip codes. A master dataset was constructed by joining four public sources: Zillow home price data (2019–2026), U.S. Census Bureau median household income, National Center for Education Statistics (NCES) public school counts, and Federal Reserve (FRED) 30-year fixed mortgage rates. Four historical 3-year rolling windows were defined, yielding **20,608 observations** across **4,782 unique zip codes** in six states (CA, TX, FL, WA, CO, AZ). Four machine learning models were trained and compared using a GroupShuffleSplit 80/20 train-test split by zip code to prevent leakage. Gradient Boosting achieved the best performance: $R^2 = 0.7025$ and MAE = 8.29%, a **332% improvement** over the single-window baseline ($R^2 = 0.1625$). Five clean features were used after removing originals and leaking columns per instructor feedback. Feature importance analysis revealed that Mortgage_Rate accounts for **65.6%** of total importance after removing Mortgage_Rate_Delta. The same Frederick, CO neighborhood returns 34.16% ROI at 2.96% rates and 2.35% at 6.81% rates, confirming that macroeconomic rate conditions are the primary driver of real estate ROI variation.

I. INTRODUCTION AND BACKGROUND

Real estate investment decisions are traditionally guided by neighbourhood-level characteristics: property prices, local income levels, and educational infrastructure. However, these factors fail to account for macroeconomic conditions that determine the affordability and demand environment in which homes appreciate or stagnate.

This project was motivated by a fundamental observation. Two homes with identical neighbourhood characteristics can produce drastically different 3-year returns depending solely on the mortgage rate at time of purchase. A buyer in 2021 at a 2.96% rate earns 34.16% ROI over three years. A buyer of the same property in 2023 at 6.81% earns only 2.35%. Same street, same price, same schools — only the rate differed.

The original version of this project studied a single time period (2023–2026) and achieved $R^2 = 0.1625$. Instructor feedback identified two key gaps: (1) interest rates were not included, and (2) only one time window was analysed. This revision addresses both by implementing a multi-window historical approach spanning four distinct economic eras.

A. Research Questions

- I. Do mortgage interest rates dominate neighbourhood factors in predicting 3-year ROI?
- II. How do socioeconomic and educational features perform when interest rates are included?
- III. Does the model generalise to unseen zip codes in different rate environments?

II. RELATED WORK

Breiman [1] introduced Random Forests, demonstrating that ensemble methods produce robust predictions and reliable feature importance rankings — justifying both the primary algorithm and importance interpretation in this study.

Case and Shiller [2] established the foundational relationship between local economic conditions and residential real estate price dynamics, providing theoretical justification for including income as a predictive feature.

Limsombunchai [3] demonstrated that non-linear models outperform traditional regression for house price prediction, directly motivating the ensemble approach over linear regression.

Gu et al. [4] applied gradient boosting to real estate valuation, finding that ensemble methods consistently outperform single models and that school proximity significantly influences property value, aligning with the inclusion of `School_Count`.

Chiarazzo et al. [5] demonstrated that educational infrastructure indicators are significant predictors of property appreciation, supporting the hypothesis that school density reflects community stability.

Freddie Mac [6] publishes the Primary Mortgage Market Survey, the authoritative source for 30-year fixed mortgage rates used in this study. Prior research consistently identifies interest rates as a primary driver of housing market activity and price appreciation cycles.

III. PROBLEM STATEMENT

The central question is: can 3-year residential real estate ROI be predicted from neighbourhood socioeconomic indicators and mortgage interest rates across multiple economic time periods?

The correlation matrix confirms non-linearity: `Mortgage_Rate` vs. `ROI` = -0.687 (strong), while `School_Count` vs. `ROI` = -0.07 (near zero). No single variable explains ROI through a linear formula.

ROI is calculated as a point-to-point 3-year return:

$$\text{ROI} = \frac{\text{Price}_{\text{exit}} - \text{Price}_{\text{entry}}}{\text{Price}_{\text{entry}}} \times 100\% \quad (1)$$

This is the standard metric used by real estate investors, representing the actual return if a property is purchased at the window start and held for exactly three years.

IV. METHODOLOGY

A. Data Sources

TABLE I
DATA SOURCES USED IN THE STUDY

Source	Coverage	Contents	Role
Zillow ZHVI	2019–2026	Monthly home prices by zip	Calculate ROI
U.S. Census ACS	2024 5-Yr	Median household income	Socioeconomic
NCES ELSI	2024–25	Public school locations	School density
FRED / Freddie Mac	2019–2026	30-yr fixed rate by year	Rate feature

B. Multi-Window Historical Approach

Four 3-year rolling windows replace the single 2023–2026 period (Table II). In the single-window approach, the mortgage rate was constant for all rows (correlation with ROI = 0.000). In the multi-window approach, the rate varies (std = 1.55), enabling it as a genuine predictive feature.

TABLE II
FOUR HISTORICAL WINDOWS

Window	Entry Rate	Δ Rate	Mean ROI	Rows
Pre-COVID 2019→2022	3.94%	+1.40%	39.46%	5,152
COVID Boom 2020→2023	3.11%	+3.70%	40.83%	5,152
Low-Rate 2021→2024	2.96%	+3.76%	28.15%	5,152
High-Rate 2023→2026	6.81%	−0.16%	0.91%	5,152
Total	2.96–6.81%	—	—	20,608

C. Feature Engineering

Five clean features are used after removing originals per instructor feedback (Table III). Two additional features were excluded: `Price_Momentum` was removed because it equals `ROI/100` (target leakage), and `Mortgage_Rate_Delta` was removed per instructor feedback.

TABLE III
FINAL FEATURE SET — ORIGINALS REMOVED AFTER ENGINEERING

Feature	Type	Purpose
<code>School_Count</code>	Original	Community stability
<code>Mortgage_Rate</code>	Original	Rate at window entry
<code>Log_Income</code>	Engineered	$\log_{1p}(\text{Median_Income})$
<code>Log_Price</code>	Engineered	$\log_{1p}(\text{Current_Price})$
<code>Price_Income_Ratio</code>	Engineered	$\text{Price} \div \text{Income}$
<i>Removed: Median_Income, Current_Price, Entry_Price</i>		
<i>Excluded: Price_Momentum (leakage), Mortgage_Rate_Delta (instructor feedback)</i>		

D. ROI Category Justification

The four ROI tier boundaries are defined using financial benchmarks:

- **Negative** ($< 0\%$): Capital loss. Zero is the universal financial breakeven.
- **Low** (0–5%): Below U.S. average inflation ($\approx 3\%/yr \approx 9\%$ over 3 years). Real purchasing power lost despite positive return.
- **Solid** (5–15%): Beats inflation. At the upper end, comparable to Treasury bonds.
- **High** ($> 15\%$): Exceeds $4.8\%/yr$ — comparable to the long-term U.S. stock market ($\approx 7\%/yr$).

E. Hyperparameter Tuning

`RandomizedSearchCV` tested 20 random combinations of five hyperparameters with 5-fold cross-validation (Table IV).

TABLE IV
HYPERPARAMETER SEARCH AND BEST CONFIGURATION

Parameter	Values Tested	Best
<code>n_estimators</code>	100, 200, 500, 1000	500
<code>max_depth</code>	None, 10, 20, 30	30
<code>min_samples_split</code>	2, 5, 10	5
<code>min_samples_leaf</code>	1, 2, 4	2
<code>max_features</code>	<code>sqrt</code> , <code>log2</code> , <code>None</code>	<code>None</code>
Best CV MAE	8.916% (5-fold)	

F. Train-Test Split

An 80/20 split by zip code using `GroupShuffleSplit` prevents data leakage across windows. No zip code appears in both training (16,464 rows, 3,825 zips) and test sets (4,144 rows, 957 zips).

V. DATA DESCRIPTION

A. Dataset Overview

TABLE V
DATASET PROPERTIES

Property	Value
Total rows	20,608 (4 windows $\times$ 5,152)
Unique zip codes	4,782
States	CA (1,511) TX (1,304) FL (914) WA (473) CO (390) AZ (288)
Rate range	2.96% to 6.81% (std = 1.55)
Train/Test	16,464 / 4,144 rows (by zip code)
ROI range	Approx. -30% to $+80\%$ across 4 eras

B. Distribution Analysis

ROI distribution varies dramatically across windows (mean 39.46% in COVID Boom vs. 0.91% in High-Rate Era). Home price shows severe right skew (skewness ≈ 4.4), requiring log transformation. Income is moderately skewed (skewness ≈ 1.3), also corrected.

C. Outlier Detection

The IQR method identified 4–6% of observations as extreme per variable. All outliers were retained — they represent genuine market events (economic collapses, luxury markets), not data entry errors.

D. Data Imbalance Assessment

The data *is* imbalanced (confirmed by instructor). Root cause: three of four windows were high-ROI boom periods, filling the High tier. Only the High-Rate Era window produced low returns.

TABLE VI
ROI TIER DISTRIBUTION — FULL DATASET (20,608 ROWS)

Tier	Count	Share	Test MAE	Status
Negative ($< 0\%$)	—	—	8.46%	Underrepresented
Low (0–5%)	—	—	4.03%	Smallest tier
Solid (5–15%)	—	—	9.37%	Hardest to predict
High ($> 15\%$)	2,922 test	dominant	8.54%	Most reliable
Overall	4,144 test	9.64:1	8.29%	Imbalanced

SMOTE does not apply — this is a regression problem. Mitigation: per-tier MAE evaluation (Table VI) quantifies where the model is most and least reliable.

VI. DATA ANALYSIS

A. Correlation Analysis

TABLE VII
KEY CORRELATIONS — MULTI-WINDOW DATASET

Feature Pair	r	Meaning
Mortgage_Rate vs. ROI	-0.687	Strong negative — was 0.000 single-window
School_Count vs. ROI	-0.070	Near zero — rate dominates across eras
Median_Income vs. Price	0.700	Wealthy areas have expensive homes

B. Model Performance Comparison

TABLE VIII
MODEL COMPARISON — ACTUAL RESULTS

Model	R^2 Test	R^2 CV-5	MAE	RMSE
Linear Regression	0.4834	0.036	11.24%	13.74%
Ridge Regression	0.4834	0.036	11.24%	13.74%
Random Forest	0.6827	0.733	8.29%	11.31%
Gradient Boosting	0.7025	0.733	8.29%	11.13%
Old single-window baseline	0.1625	—	5.75%	—

Gradient Boosting achieves $R^2 = 0.7025$, a **332% improvement** over the baseline. MAE increased from 5.75% to 8.29% because the ROI range across four eras spans approximately 110 percentage points, making the prediction task substantially harder. The R^2 improvement is the correct measure of advancement.

C. 5-Fold Cross-Validation

TABLE IX
5-FOLD CROSS-VALIDATION — GRADIENT BOOSTING

Fold	MAE	R^2
Fold 1	7.91%	0.7364
Fold 2	8.14%	0.7259
Fold 3	8.02%	0.7305
Fold 4	8.05%	0.7334
Fold 5	7.85%	0.7397
Mean	$8.00 \pm 0.10\%$	0.7332 ± 0.0047

The MAE spread of 0.29% across folds confirms model stability.

D. Feature Importance

After removing Mortgage_Rate_Delta per instructor feedback and renormalising the remaining five features to 100%:

TABLE X
FEATURE IMPORTANCE — 5 FEATURES (DELTA REMOVED)

Rank	Feature	Importance	Interpretation
1	Mortgage_Rate	65.6%	Was 0% single-window; dominant signal
2	Price_Income_Ratio	11.2%	Affordability — undervalued markets grow more
3	School_Count	10.4%	Was #1 single-window; community stability
4	Log_Price	7.3%	Entry price (log-scaled)
5	Log_Income	5.5%	Neighbourhood income (log-scaled)

Mortgage_Rate alone explains 65.6% of all ROI variation — nearly two thirds. School_Count dropped from #1 (single-window) to #3 (multi-window) because the rate signal is substantially larger. Schools still matter as a community stability indicator but operate within the macroeconomic rate environment.

VII. EXPECTED OUTCOMES

Four outcomes were anticipated at project start, all achieved:

- 1) **Accurate prediction:** $R^2 = 0.7025$, CV $R^2 = 0.7332 \pm 0.0047$ — strong and stable.
- 2) **Feature ranking:** Mortgage_Rate at 65.6% is the primary finding.
- 3) **Investment tool:** `check_investment()` delivers Solid, Caution, or Hidden Gem verdicts.
- 4) **Real-world validation:** Frederick, CO tested in two rate environments confirms rate-driven behaviour.

The most significant unexpected finding: the 332% improvement in R^2 from adding historical rate data demonstrates that the original project was missing the dominant signal entirely.

VIII. REAL-WORLD VALIDATION — FREDERICK, CO

Frederick, Colorado was tested with identical neighbourhood inputs (\$75,000 income, 12 schools, \$142,300 price) in two rate environments:

TABLE XI
FREDERICK CO VALIDATION — TWO RATE ENVIRONMENTS

Scenario	Rate	Pred. ROI	Range	Verdict
2021 (low-rate)	2.96%	34.16%	25.88%–42.44%	SOLID
2023 (high-rate)	6.81%	2.35%	–5.93%–10.63%	CAUTION

Same city, same income, same schools, same price. The rate changed from 2.96% to 6.81%. The predicted return changed from 34.16% to 2.35% — a difference of 31.81 percentage points driven entirely by one variable.

IX. PROJECT TIMELINE

TABLE XII
PROJECT EXECUTION TIMELINE

Period	Activity
Weeks 1–2	Data acquisition: Zillow, Census, NCES, FRED
Week 3	Cleaning: GEO_ID → zip, joins, normalisation
Week 4	Multi-window framework; rate features added
Week 5	EDA: distributions, correlations, imbalance
Week 6	Feature engineering, split, hyperparameter tuning
Week 7	4-model comparison, CV, feature importance
Week 8	Validation, report (IEEEtran), slides, submission

X. CONCLUSION

This project demonstrated that historical mortgage interest rates are the dominant driver of residential real estate ROI across economic eras. Using four historical windows spanning 2019–2026, R^2 improved 332% from 0.1625 to 0.7025.

After removing `Mortgage_Rate_Delta` per instructor feedback, `Mortgage_Rate` alone accounts for 65.6% of feature importance. Frederick, CO returns 34.16% ROI at 2.96% rates and 2.35% at 6.81% rates — same neighbourhood, different eras.

Five-fold cross-validation confirms stability ($\text{MAE} = 8.00 \pm 0.10\%$, $R^2 = 0.7332 \pm 0.0047$). Data imbalance (9.64:1, High vs. Low) is documented honestly with per-tier MAE evaluation.

A. Limitations

- Imbalance: 3 of 4 windows are boom periods. Adding 2015–2018 windows would balance Low/Negative tiers.
- Geographic scope: 6 states only. Expanding to all 50 states improves generalisability.
- School count vs. quality: count used as proxy; quality ratings not publicly available at zip code scale.

B. Future Work

- Add 2015→2018 window for Low/Negative tier balance
- Expand to all 50 U.S. states
- Add school quality ratings and crime rates
- Live market feeds for real-time forecasting
- Mobile application delivering investor verdicts

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